

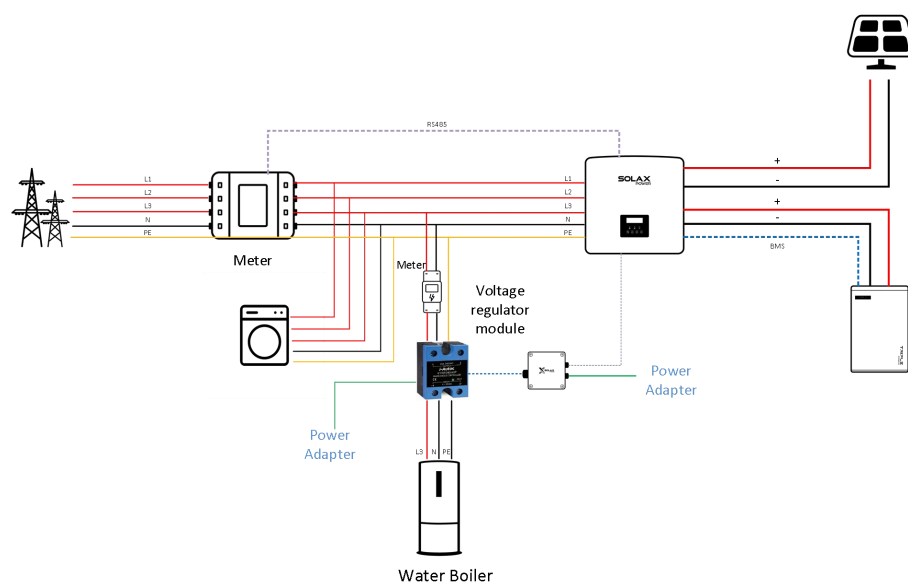
# How to Use a Water Boiler in the Adapter Box G2 System



## Declaration

In some European regions, such as Austria and Germany, there is a greater preference for using more economical and equally durable water boilers compared to heat pumps. The SolaX Power adapter box G2 is designed for smart energy management of heat pumps within inverter systems. For water boilers that support AI (analog input) control, the Adapter Box G2 has achieved compatibility and control with them.

The system includes an energy storage system or an on-grid inverter system, an adapter box G2, a pressure regulation module (to be provided by the customer), and a water boiler.



The compatible inverter versions supported by this solution are:

### On-grid Inverter:

- X3-PRO G2
- X1-BOOST G4
- X3-MIC G2
- X1-Mini G4

### Hybrid Inverter:

- X1-Hybrid-G4
- X3-Ultra
- X3-Hybrid-G4
- X3-IES

The firmware version requirement for the Adapter Box G2 in this solution is: **007.01**

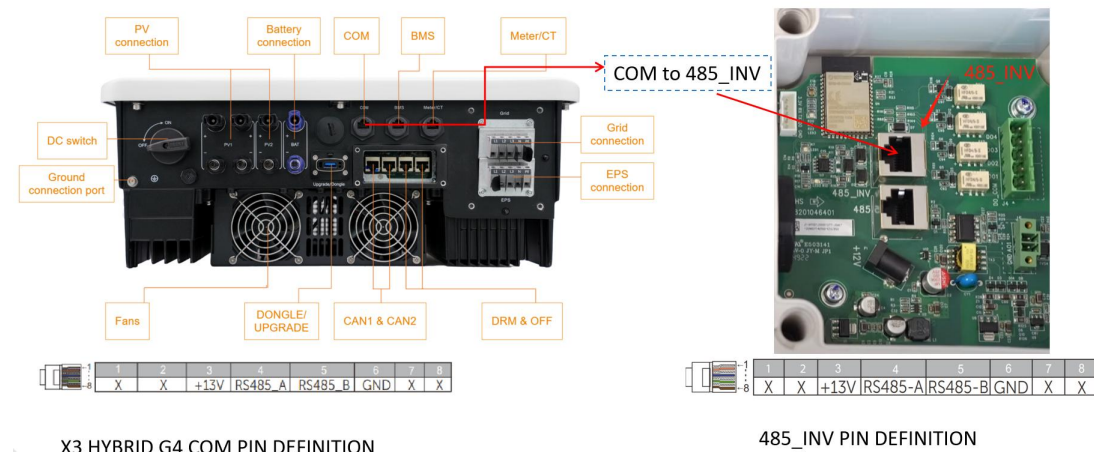
## Preparation

## Step 1

### Connect the Adapter Box G2 to the inverter

The Adapter Box G2 needs to obtain grid information and current PV power from the inverter to calculate the excess PV power available for the water boiler. If this system is used in areas without power limitation, the inverter can be replaced by a meter. Additionally, this solution can be applied to other inverter systems (including third-party inverters).

Taking the X3-Hybrid G4 as an example:



### Note:

If the inverter's communication port does not include PIN3 and PIN6 with 13V DC, then the Power Adapter for the Adapter Box G2 needs to be connected.

## Step 2

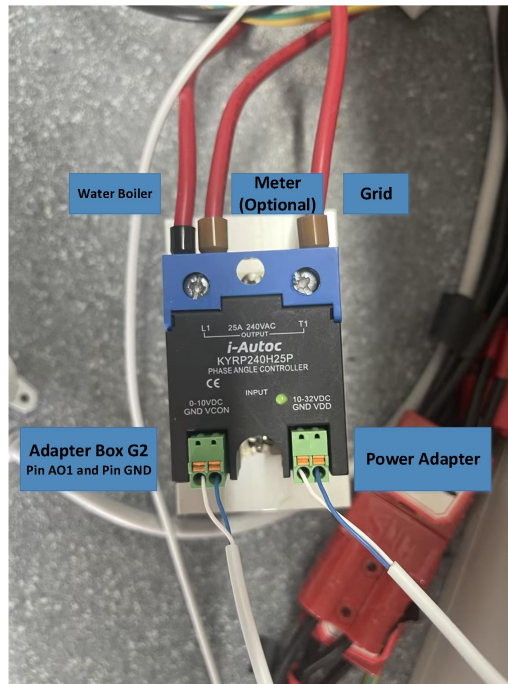
### Connect the Adapter Box G2 to the voltage regulator module

Connect the AI port of the Adapter Box G2 to the voltage regulator module, and connect the voltage regulation module to the grid. It is important to note that the single-phase meter connected to the voltage regulation module is used to provide feedback to the system on the actual operating power of the water boiler. If not needed, it can be removed from the system.

### Operation Logic

The voltage regulation module will determine the current output power of the water boiler based on the voltage value of the analog signal output by the Adapter Box G2.

*For example, if the analog signal is 5V and the rated power of the water boiler is 5kW, the voltage regulation module will supply 2.5kW of power to the water boiler.*



### Step 3

#### Connect the Water Boiler to the voltage regulator module

It is recommended that customers use a power strip for this system. Cut the power strip's cable and connect it to the output port of the voltage regulator module, then connect the water boiler to the power strip. Alternatively, the water boiler's cable can be cut and connected directly to the voltage regulation module.

### Step 4

#### Configuration

Set the control type of Adapter Box G2 as Analog output. Then, based on the operating logic and rated power of the Water Boiler, set the 'Max Step' and 'Load Capacity' on the Adapter Box G2. You can either use the local function on SolaXCloud by scanning the QR code on the Adapter Box G2 for local setup or configure it remotely on SolaXCloud."

*Path:*

*Settings>>Control type>>Analog output*

*Settings>>Max step*

*Settings>>Load capacity*

<

Settings

Control type

Analog output ▾

Max step

7

15

Load capacity (W)

6000

Save